[METRIC]

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SUPERSEDING
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2 October 1967

MILITARY SPECIFICATION

CALCIUM OXIDE, TECHNICAL (METRIC)

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two grades of technical calcium oxide.

1.2 Classification. Calcium oxide shall be of the following grades as specified (see 6.2):

Grade A - 90 percent purity
Grade B - 95 percent purity

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

FSC 6810

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS (CFR)

40 CFR 171 to 179 - Department of Transportation Hazardous Materials Regulations

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders for the above publication should cite "49 CFR 171 to 179.")

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

ASTM STANDARDS

C25 - Chemical Analysis of Limestone, Quicklime, and Hydrated Lime
D1193 - Reagent Water
E11 - Wire-Cloth Sieves for Testing Purposes

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Chemical and physical characteristics. Calcium oxide shall conform to the applicable chemical and physical characteristics of table I when tested as specified therein.

TABLE I. Chemical and physical characteristics

Characteristic	Percent by weight				Test paragraph
	Grade A		Grade B		
	Min	Max	Min	Max	
Nonvolatile matter	94.0		98.0		4.2.4.1
Calcium oxide*	90.0		95.0		4.2.4.2
Magnesium oxide*				1.0	4.2.4.3
Iron and aluminum oxides*				1.0	4.2.4.4
Silicon dioxide*				1.5	4.2.4.5
Particle size:					4.2.4.6
Retained on 38.1-mm square openings		0.0			
Retained on 25.4-mm square openings		8.0			
Retained on No. 1 sieve	95.0				
Retained on No. 3-1/2 sieve				0.0	
Retained on No. 12 sieve			95.0		

*On nonvolatile basis

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the calcium oxide of one grade produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

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4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test. Sampling shall be conducted in accordance with table II. A representative specimen of approximately 250 grams (g) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE II. Sampling for test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled shipping container, ready for shipment. Sample shipping containers shall be examined for the following defects grouped collectively using an AQL of 2.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Marking incorrect, missing, or illegible
(f) Unitization not as specified

4.2.3.2 For test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Nonvolatile matter. Weigh to the nearest 0.01 g approximately 1 g of the specimen into a tared platinum crucible. Cover with a lid, place in an electric muffle furnace, and heat gently for 5 minutes. Gradually increase the

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temperature of the furnace until the temperature reaches approximately 1000°C. Maintain this temperature until constant weight is obtained (usually 15 minutes). Calculate the percent by weight nonvolatile matter as follows:

$$\text{Percent nonvolatile matter} = \frac{100A}{W}$$

where: A = Weight of specimen after ignition in grams and
W = Initial weight of specimen in grams.

4.2.4.2 Calcium oxide. Determine the percent by weight calcium oxide in the specimen in accordance with the applicable test procedure in ASTM C25. Calculate the percent by weight calcium oxide on a nonvolatile basis as follows:

$$\text{Percent calcium oxide on a nonvolatile basis} = \frac{100A}{B}$$

where: A = Percent calcium oxide and
B = Percent nonvolatile matter determined in 4.2.4.1.

4.2.4.3 Magnesium oxide. Determine the percent by weight magnesium oxide in the specimen in accordance with the applicable test procedure in ASTM C25. Calculate the percent by weight magnesium oxide on a nonvolatile basis as follows:

$$\text{Percent magnesium oxide on a nonvolatile basis} = \frac{100A}{B}$$

where: A = Percent magnesium oxide and
B = Percent nonvolatile matter determined in 4.2.4.1.

4.2.4.4 Iron and aluminum oxides. Determine the percent by weight iron and aluminum oxides in the specimen in accordance with the applicable test procedure in ASTM C25. Calculate the percent by weight iron and aluminum oxides on a nonvolatile basis as follows:

$$\text{Percent iron and aluminum oxides on a nonvolatile basis} = \frac{100A}{B}$$

where: A = Percent iron and aluminum oxides and
B = Percent nonvolatile matter determined in 4.2.4.1.

4.2.4.5 Silicon dioxide. Determine the percent by weight silicon dioxide in the specimen in accordance with the applicable test procedure in ASTM C25. Calculate the percent by weight silicon dioxide on a nonvolatile basis as follows:

$$\text{Percent silicon dioxide on a nonvolatile basis} = \frac{100A}{B}$$

where: A = Percent silicon dioxide and
B = Percent nonvolatile matter determined in 4.2.4.1.

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4.2.4.6 Particle size. Tare and nest the applicable sieves specified in table I and conforming to ASTM E11 in order of increasing fineness with the coarsest sieve on top and place on a bottom pan. Weigh to the nearest 0.01 g approximately 100 g of specimen and transfer to the top sieve. Cover the top sieve and shake for 5 minutes on a mechanical shaker geared to produce 300 ± 15 gyrations and 150 ± 10 taps of the striker per minute. Weigh the amount of specimen retained on each sieve and calculate the percent by weight of specimen retained on each sieve.

5. PACKAGING

5.1 Preservation. Calcium oxide shall be preserved level A, level B, or commercially as specified (see 6.2).

5.1.1 Level A. A quantity of 36.3 (+0.4 or -0) kilograms of calcium oxide shall be unit packed in a removable head steel drum conforming to Department of Transportation (DOT) Specification 17C, 17H, or 37A (see 6.5). The drum shall be closed in accordance with the drum manufacturer's directions.

5.1.2 Level B. A quantity of 36.3 (+0.4 or -0) kilograms of calcium oxide shall be unit packed in a waterproof, multiwall paper sack conforming to DOT Specification 44B or 44D or in a waterproof, laminated textile shipping bag conforming to DOT Specification 36B (see 6.5). The bag shall be closed in accordance with the bag manufacturer's directions.

5.1.3 Commercial. Calcium oxide shall be unit packed to assure maintenance of specified purity and to assure carrier acceptance and safe delivery to the first domestic destination. Containers shall be in compliance with DOT regulations and Uniform Freight Classification Rules, National Motor Freight Classification Rules, or the rules of other common carriers applicable to the mode of transportation.

5.2 Packing. Calcium oxide, unit packed as specified in 5.1, shall require no further protection for shipment.

5.3 Unitization. Packs of calcium oxide shall be unitized in accordance with the applicable palletization requirements of MIL-STD-147.

5.4 Marking. Packs of calcium oxide shall be marked in accordance with MIL-STD-129, DOT regulations, and any other applicable regulatory requirements. In addition, each pack shall be marked to show the lot or batch number, the date of manufacture of the calcium oxide, and the following precautionary information:

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HAZARD: Corrosion, Eye and Skin

DANGER! CAUSES SEVERE BURNS

Do not get in eyes, on skin, or on clothing.
Avoid breathing dust.
Keep container closed.
Use with adequate ventilation.
Wash thoroughly after handling.
When handling, wear goggles, face shield, and gloves.

FIRST AID: In case of contact, immediately flush eyes or skin with plenty of water for at least 15 minutes while removing contaminated clothing and shoes. Call a physician. Wash clothing before reuse. Discard contaminated shoes. If swallowed and victim is conscious, have victim drink water or milk. Do not induce vomiting.

6. NOTES

6.1 Intended use. Calcium oxide is intended for use in steel furnace fluxing, in neutralizing acidic wastes, and in manufacturing bleaching material.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Grade of calcium oxide required (see 1.2)
(c) Level of preservation required (see 5.1)

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

6.5 Container outage. A minimum of 30 percent added outage volume above that of the calcium oxide should be allowed per container to provide for expansion of contents due to conversion of the oxide to hydroxide and carbonate of reduced density during long periods of storage.

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